SIRP: -01 Update — Identity-Aware Routing

Adding User Identity to Content-Driven AI Routing

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draft-chen-nmrg-semantic-inference-routing-01

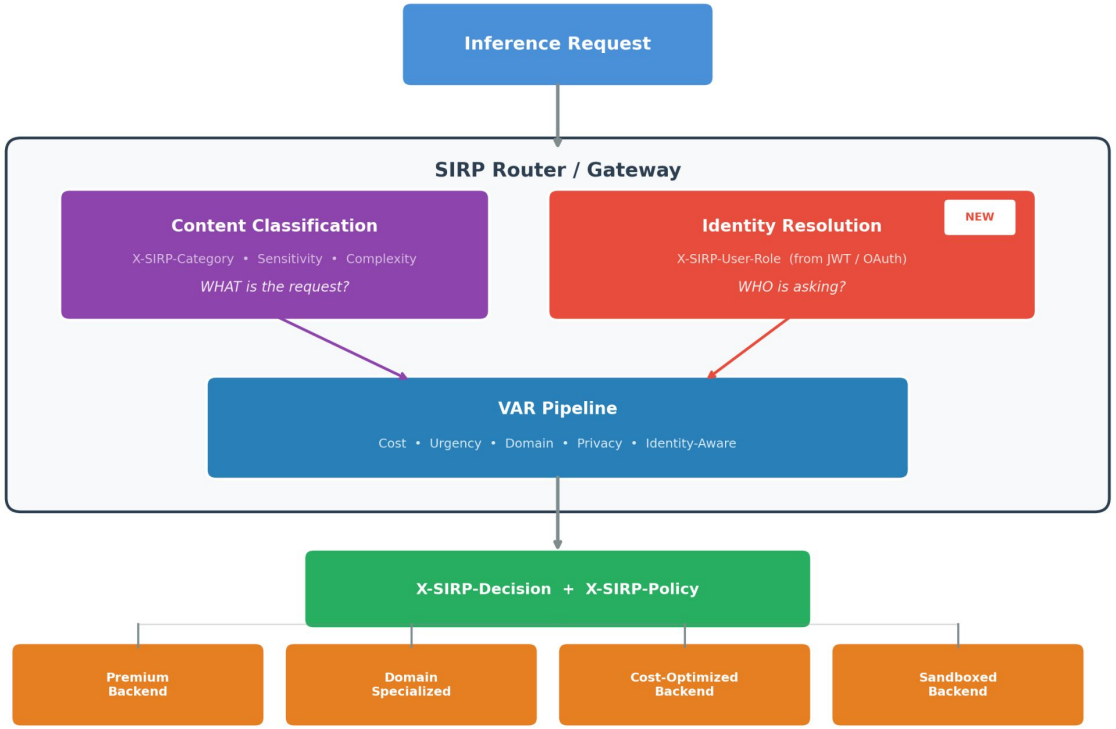
Recap: What -00 Established

- ✅ Content-level classification for routing (not metadata-only)
  - X-SIRP-Category (math, code, business, adversarial...)
  - X-SIRP-Sensitivity (low | medium | high)
  - X-SIRP-Complexity (1-5 reasoning effort)
- ✅ VAR modules: cost, urgency, domain specialization, privacy
- ✅ Reference impl: vllm-project/semantic-router
- 🔴 Gap in -00:
  - Routing only considers WHAT is asked, not WHO is asking
  - Cannot enforce per-tenant model access or usage policies

What's New in -01: Two-Dimensional Routing

- 📐 Content dimension (existing):
  - What is the request? → Category, Sensitivity, Complexity
- 👤 Identity dimension (NEW):
  - Who is asking? → X-SIRP-User-Role
  - Resolved from upstream auth (JWT, OAuth)
  - Values: admin, premium, free, etc.
- 🔗 Combined routing:
  - Same "code" request → premium model (admin)
  - Same "code" request → lite model (free-tier)
- ⚙️ Fits naturally as a new VAR module
  - No structural changes to SIRP pipeline

New Header: X-SIRP-User-Role



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New VAR Module: Identity-Aware Routing

- 📋 Existing VAR pipeline (unchanged):
  - Cost Optimization → Urgency → Domain Specialization → Privacy

- ➕ New module: Identity-Aware Routing
  - Combines X-SIRP-User-Role with classification results
  - Admin + high-complexity → premium model pool
  - Free + low-complexity → cost-optimized model
  - Role bindings and permitted backends are configurable

- 🔗 Composable with other VAR modules:
  - identity + domain: admin code request → premium code backend
  - identity + privacy: free user + high sensitivity → sandboxed lite
  - identity + urgency: premium user + urgent → fast premium backend

Example: Identity-Aware Code Routing

- 📝 Input: "Optimize this SQL query for large-scale analytics."
- 🔍 Classification:
  - Category: code | Sensitivity: low | Complexity: 3
- 👤 Admin (X-SIRP-User-Role: admin from JWT):
  - ✅ Decision: code-premium-v2
  - ✅ Policy: identity-admin,domain-code
  - ✅ Cost optimization: skipped (admin override)
- 👤 Free-tier (X-SIRP-User-Role: free):
  - ✅ Decision: code-lite-v1
  - ✅ Policy: identity-free,domain-code,low-cost
  - ✅ Cost optimization: applied

-01 Summary & Next Steps

- 📊 Revision scope: 8.3% growth (672 → 728 lines) — very lean
- ✨ Key additions:
  - X-SIRP-User-Role header (one new header)
  - Identity-Aware Routing VAR module (one new module)
  - One new example + security considerations
  - New reference: RFC 7519 (JWT)
- 📍 Next steps:
  - Seeking feedback on identity-content interaction model
  - Rate limiting as potential future VAR module
  - Integration with multi-provider API draft (shared authz model)
  - Complements draft-chen-nmrg-multi-provider-inference-api-01

Multi-Provider AI Inference API: -01 Update

Authorization, RBAC & Rate Limiting for Multi-Tenant Deployments

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Recap: What -00 Established

- ✅ OpenAI Responses API as de facto standard interface
- ✅ "auto" parameters for vendor-neutral model/tool/reasoning selection
- ✅ Extension headers (X-AI-*) for routing hints & transparency
- ✅ Multi-provider orchestration: failover, streaming, workflows
- ✅ Security-aware and performance-based auto-selection

- 🔴 Gap identified in -00:
  - WHO is making the request was ignored
  - No user authentication, role-based access, or usage governance
  - Production multi-tenant deployments need identity-aware routing

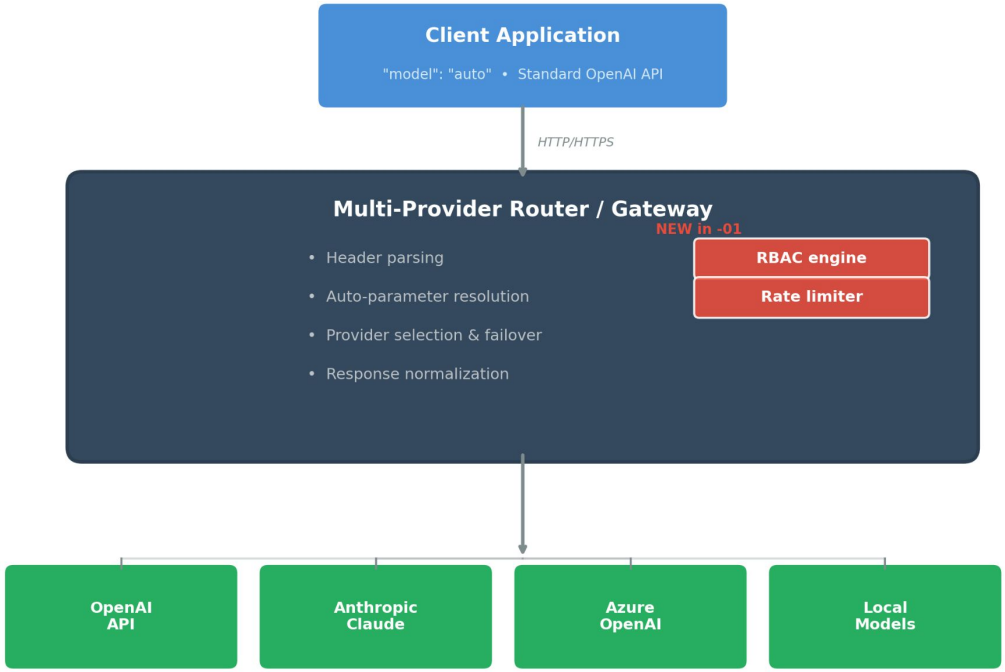
What's New in -01: Three Additions to Extension Headers

- 1️⃣ Authorization & Identity Headers
  - X-Authz-User-Id, X-Authz-User-Groups, X-Authz-User-Roles
  - Injected by upstream auth gateways (JWT, OAuth 2.0)

- 2️⃣ RBAC Framework
  - Role bindings → routing decisions → priority ordering
  - Admin → premium models | Free → lite models

- 3️⃣ Rate Limiting
  - X-RateLimit-* and X-TokenLimit-* response headers
  - RPM (requests) + TPM (tokens) per user/group/model
  - External (Envoy RLS) or local (sliding window) providers

Updated Architecture



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Example: RBAC-Aware Routing

- 📝 Same request, different users:

- 👤 Admin (X-Authz-User-Groups: platform-admins)
  - → X-AI-RBAC-Role: admin
  - → X-AI-Model-Mapped: qwen-14b-instruct
  - → Reasoning: enabled, full model pool

- 👤 Free-tier (X-Authz-User-Groups: free-tier)
  - → X-AI-RBAC-Role: free_user
  - → X-AI-Model-Mapped: qwen-7b-instruct
  - → Reasoning: disabled, limited model pool

- ✅ Content classification + identity = richer routing decisions
- ✅ RBAC policies combine with task hints for context-aware authz

Example: Rate Limiting in Action

- 🚫 Rate limit exceeded (429):
  - X-RateLimit-Limit: 100
  - X-RateLimit-Remaining: 0
  - X-RateLimit-Retry-After: 42 seconds

- ✅ Successful request with budget headers:
  - X-RateLimit-Limit: 1000 | X-RateLimit-Remaining: 847
  - X-TokenLimit-Limit: 1M | X-TokenLimit-Remaining: 345K

- ⚙️ Implementation modes:
  - Fail-closed (default): reject on limiter error → safe
  - Fail-open: allow on limiter error → available
  - Hybrid chain: external (Envoy RLS) + local fallback

Security Considerations (-01 Additions)

- 🔒 Identity Header Security
  - X-Authz-* headers MUST only be accepted from trusted auth gateways
  - Router SHOULD strip these from client requests
  - Prevents authentication bypass attacks

- 🔒 RBAC Policy Security
  - Role binding configs need access controls
  - Misconfigured RBAC → unauthorized access to premium models

- 🔒 Rate Limit Bypass Prevention
  - Fail-closed mode recommended for production
  - Rate limiter failures reject requests (not allow)

- ✅ All extensions remain OPTIONAL — zero impact on unaware clients

-01 Summary & Next Steps

- 📊 Revision scope: 11.4% growth (2464 → 2744 lines)
- ✨ Key additions:
  - Authorization headers (JWT/OAuth integration)
  - RBAC framework (role bindings → routing decisions)
  - Rate limiting (RPM/TPM per user/group/model)
  - 3 new references: RFC 6750, RFC 7519, Envoy RLS
- 📍 Next steps:
  - Seeking feedback on header naming conventions
  - Experimental deployments with real RBAC policies
  - Integration with SIRP for semantic + identity routing
  - Complements draft-chen-nmrg-semantic-inference-routing-01

Questions?

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